

INVARIANT IDEALS AND THE HOVEY–STRICKLAND CONJECTURE

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ABSTRACT. We prove that the radical ideals of a Lubin–Tate deformation ring stable under an open subgroup of the extended Morava stabilizer are precisely the height ideals and the unit ideal. This classification implies the Hovey–Strickland conjecture at every prime and height. The proof combines compatible finite Honda actions, a crystalline exclusion theorem for equivariant formal subgroups, and the Gross–Hopkins period map.

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1. INTRODUCTION

We prove the Hovey–Strickland conjecture at every prime and height. The main arithmetic result classifies radical ideals of Lubin–Tate deformation rings preserved by open subgroups of the extended Morava stabilizer.

Invariant ideals. Fix a prime p and an integer $n \geq 1$. Let D_n be the central division algebra over $\mathbf{Q}_p$ of invariant $1/n$, and let

$$k_n = \mathbf{F}_{p^n}, \quad R_n = W(k_n)[[u_1, \dots, u_{n-1}]], \quad S_n = \mathcal{O}_{D_n}^\times.$$

The extended Morava stabilizer is $\mathbb{G}_n = S_n \rtimes \text{Gal}(k_n/\mathbf{F}_p)$. Its action on R_n is the marked action on the universal deformation. We use the height ideals

$$I_0 = (0), \quad I_h = (p, u_1, \dots, u_{h-1}) \quad (1 \leq h \leq n), \quad I_{n+1} = R_n.$$

The ideals I_h are invariant under $\mathbb{G}_n$.

Theorem 1.1. *Let H be an open subgroup of $\mathbb{G}_n$.*

- (i) *The H -stable radical ideals of R_n are $I_0, \dots, I_{n+1}$.*
- (ii) *The proper prime ideals of R_n with open stabilizer in $\mathbb{G}_n$ are $I_0, \dots, I_n$.*

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Thick tensor ideals. Let $\mathcal{C}_n$ denote the rigid tensor-triangulated category of dualizable $K(n)$ -local spectra, with its completed tensor product. For a finite p -local spectrum $F(j)$ of type j , set

$$\mathcal{D}_j = \text{thick}_{\otimes}(L_{K(n)}F(j)) \quad (0 \leq j \leq n), \quad \mathcal{D}_{n+1} = (0).$$

Here type j means that the first nonzero Morava K -theory of the finite spectrum occurs at height j .

Theorem 1.2. *For every prime p and every height $n \geq 1$:*

(i) *The thick tensor ideals of $\mathcal{C}_n$ form the chain*

$$\mathcal{C}_n = \mathcal{D}_0 \supsetneq \mathcal{D}_1 \supsetneq \cdots \supsetneq \mathcal{D}_n \supsetneq \mathcal{D}_{n+1} = (0).$$

(ii) *For $0 \leq j \leq n$, the ideal $\mathcal{D}_j$ is independent of the choice of the finite spectrum of type j .*

(iii) *The Balmer spectrum has points $\mathcal{D}_1, \dots, \mathcal{D}_{n+1}$ and specialization order*

$$\overline{\{\mathcal{D}_i\}} = \{\mathcal{D}_j : j \geq i\}.$$

With $H = \mathbb{G}_n$, Theorem 1.1(i) supplies the classification of invariant radical ideals required by [BHN22, Theorem 4.9], which gives Theorem 1.2(i). Parts (ii) and (iii) of Theorem 1.2 follow from [BHN22, Propositions 3.3 and 3.5]. At height zero, the classification follows from finite-dimensional graded rational vector spaces.

Finite orbit identities. For $1 \leq h < n$, write

$$B_h = R_n/I_h = k_n[[u_h, \dots, u_{n-1}]], \quad x = u_h.$$

Let ρ_g denote the marked action of $g \in \mathbb{G}_n$.

Corollary 1.3. *Let $H \subseteq \mathbb{G}_n$ be open.*

(i) *For $1 \leq h < n$ and $0 \neq f \in B_h$, there exist integers $N \geq 0$ and $s \geq 1$, elements $g_1, \dots, g_s \in H$, and coefficients $b_1, \dots, b_s \in B_h$ such that*

$$x^N = \sum_{i=1}^s b_i \rho_{g_i}(f).$$

(ii) *For $0 \neq f \in R_n$, there exist integers $N \geq 0$ and $s \geq 1$, elements $g_1, \dots, g_s \in H$, and coefficients $b_1, \dots, b_s \in R_n$ such that*

$$p^N = \sum_{i=1}^s b_i \rho_{g_i}(f).$$

Idea of the proof. The main step is to exclude a nonzero H -stable prime $P \subset B_h$ with $x \notin P$. Let $d = n - h$. The full p -divisible group over B_h/P has generic connected part of height h and étale quotient of height d . Its torsion coordinates, together with the Kodaira–Spencer map, give compatible actions of the finite Honda groups $\mathbb{H}_h[p^m]^d$ by Propositions 3.1 and 3.2.

After restricting H to an open subgroup of its intersection with the norm-one group, choose the preserved frame component supplied by Subsection 4.1. Let K be its field and let $\Lambda = K^H$. The equations of P and the H -invariant fractions define the relative orbit algebras

$$K \otimes_{K^{p^h m} \Lambda} K.$$

Their relations descend to compatible Hopf ideals by Lemma 4.3. The lengths and tangent dimensions of these algebras then give a formal subgroup of $\mathbb{H}_h^d$, of dimension r and height hr , with $0 < r < d$ (Proposition 4.4). The changes of frame descend this subgroup to

$$Y \subset U := T_p(G_{\text{et}})^{\vee} \otimes_{\mathbf{Z}_p} G_{\text{conn}},$$

with its inclusion preserved by the comparisons for H .

Theorem 2.1 realizes the full group over a discrete valuation ring. Lemma 6.2 provides an equivariant extension with perfect residue field and an H -fixed uniformizer. Crystalline comparison then gives a quotient

$$\mathrm{Hom}(M_{\mathrm{et}}, C) \longrightarrow Q, \quad \mathrm{rank}(M_{\mathrm{et}}) = d, \quad \mathrm{rank}(C) = h, \quad \mathrm{rank}(Q) = hr,$$

where M_{et} has slope zero and C has slope $1/h$. Propositions 6.5 and 6.6, using [Tsu23, Theorem 2.14] and [Pal22, Theorems 2.9 and 2.14], extend the smallest overconvergent hull of $(\det Q)^\vee$ across the special fibre. Frobenius descent and the marked action of H yield the rational GL_n -equivariant projector of Proposition 6.9. Lemma 6.8 identifies the extended hull with the projector's image. In the proof of Theorem 6.1, this projector forces $Q^\vee$ to be a $\mathrm{GL}(M_{\mathrm{et}}) \times \mathrm{GL}(C)$ -stable subspace of $M_{\mathrm{et}} \otimes C^\vee$. This representation is irreducible, which contradicts $0 < hr < hd$.

In characteristic zero, Lemma 7.3 uses the Gross–Hopkins period map to interpolate a small part of each stabilizer orbit by an analytic chart. Every element of a proper stable ideal vanishes along the orbit of a common zero. The chart forces its germ to vanish, giving the ideal rigidity of Theorem 7.4. This rigidity and the exclusion of P give the prime classification. The radical ideal classification follows from Lemma 8.1, which gives open stabilizers for the finitely many minimal primes of an invariant ideal. Finally, localizing an orbit ideal at x or p and clearing denominators gives the finite identities of Corollary 1.3.

Related results. Hovey and Strickland stated the conjecture in [HS99, p. 61], with its proposed ideals defined in [HS99, Definition 12.14, p. 62]. Barthel, Heard, and Naumann give the formulation in terms of thick tensor ideals in [BHN22, Conjecture 3.4]. The arithmetic question originates in Chai's study of prime ideals of R_n/p preserved by open subgroups of $\mathbb{G}_n$. The corresponding classification of $\mathbb{G}_n$ -invariant radical ideals of R_n is formulated in [BHN22, Hope 4.1 and Remark 4.2].

Barthel, Heard, and Naumann prove the Hovey–Strickland conjecture at height two for every prime [BHN22, Theorem 4.15]. Their arithmetic result classifies the associated primes of $\mathbb{G}_2$ -invariant ideals of R_2 [BHN22, Theorem 4.16]. Theorem 1.1 treats open subgroups in every height.

Overview of the contents. Section 2 establishes Frobenius coordinates for the full p -divisible group. Section 3 constructs the finite Honda actions. Sections 4 and 5 associate a formal subgroup and a valued realization to a stable prime. Section 6 proves the crystalline tensor exclusion theorem, and Section 7 treats the generic fibre through the Gross–Hopkins period map. Section 8 deduces the invariant ideal classification, the finite orbit identities, and the classification of thick tensor ideals.

2. THE FULL GROUP AND ITS FROBENIUS COORDINATES

Fix integers $1 \leq h < n$, and let $q = p^h$ and $d = n - h$. Let k be a finite field containing $\mathbf{F}_{p^n}$, and write

$$B = k[[u_h, \dots, u_{n-1}]], \quad x = u_h.$$

All stabilizer actions are those on the marked universal height- n deformation, including its comparison isomorphisms. We may restrict an open subgroup to its intersection with the group of reduced norm one. The intersection is open in the latter group. All subsequent restrictions will be open in this norm-one subgroup.

Let $\mathbb{H}_j$ denote the Honda formal group of height j , in a coordinate for which $[p]_{\mathbb{H}_j}(T) = T^{p^j}$, and let D_j be its rational endomorphism division algebra. Thus $\dim_{\mathbf{Q}_p} D_j = j^2$. If a superscript (p^s) is attached to a formal group, a series, or a morphism, it denotes the relative Frobenius twist of its coefficients. All series in Sections 2–5 are evaluated using the original full formal group law.

Theorem 2.1. *Suppose that a nonzero prime $P \subset B$ is stable under an open subgroup H of the height- n stabilizer, or of its norm-one subgroup, and that $x \notin P$. After replacing H by an open subgroup of its intersection with the norm-one group, the following data exist.*

- (i) *There are a finite extension $K_1/\text{Frac}(B/P)$ preserved by H and a p -divisible subgroup*

$$Y_1 \longrightarrow U = (T_p G_e)^\vee \otimes_{\mathbf{Z}_p} G_c, \quad \dim Y_1 = r, \quad \text{ht } Y_1 = hr, \quad 0 < r < d.$$

Here G is the original full height- n group over K_1 , and G_c, G_e are its connected part and etale quotient, of heights h, d , respectively. The inclusion is covariant for the original comparison maps.

- (ii) *The group G has a centered realization over a complete discrete valuation ring preserved by H , followed by an equivariant extension to $k'[[\pi]]$, where k' is perfect and H fixes π .*
 (iii) *The full rational Dieudonne crystal yields a good module M over $W(k')[[\pi]][1/p]$ with the original marked Honda special fibre. Over the generic Amice field $\mathcal{E}$, the group maps induce maps covariant for H ,*

$$0 \longrightarrow M_{\text{et}} \longrightarrow M_{\mathcal{E}} \longrightarrow C \longrightarrow 0, \quad \text{Hom}(M_{\text{et}}, C) \longrightarrow Q \longrightarrow 0,$$

where $M_{\mathcal{E}} = M \otimes \mathcal{E}$ and $(\text{rank } M_{\text{et}}, \text{rank } M, \text{rank } C, \text{rank } Q) = (d, n, h, hr)$. The module M_{et} has slope zero, and C, Q have slope $1/h$.

The proof of Theorem 2.1 occupies Sections 2–5.

2.1. The full torsion group. Let $R_0 = k[[u_1, \dots, u_{n-1}]]$. Over this complete deformation ring the Weierstrass kernels of $[p^m]$ are finite flat, compatible in m , and form the full height- n p -divisible group. Their algebraic finite flat models and their transition maps survive quotient and localization. The comparison between formal and algebraic groups is given by [dJ95, Lemma 2.4.4 and Proposition 2.4.8, pp. 23–26]; the Dieudonne crystal of the full group is constructed in [dJ95, Sections 2.3.3–2.3.4]. In particular, the coordinate of each original finite torsion scheme continues to generate its finite coordinate algebra after x is inverted. On the height- h stratum the full group retains height n , while its connected formal part has height h .

On that stratum the first nonzero term of the formal p -series is xT^q . There is an exact factorization

$$(2.1) \quad [p^m] = V_m \circ \text{Frob}^{hm}.$$

The Frobenius kernel is connected of rank p^{hm} , whereas $\ker V_m$ is etale of rank p^{dm} . Consequently $\ker V_m$ is the etale quotient of $G[p^m]$. Choose compatible ordered $\mathbf{Z}_p/p^m$ -bases of these kernels, and denote their coordinate values by $y_{i,m}$, $1 \leq i \leq d$. Writing $V = V_1$, compatibility means

$$(2.2) \quad y_{i,m-1} = V^{(q^{m-1})}(y_{i,m}).$$

This is the transition induced by multiplication by p on the etale Tate module. Evaluations of formal series at these coordinates are first performed in the original complete finite torsion algebra, where the coordinate is topologically nilpotent. Localization is performed after evaluation.

2.2. Frobenius bases. The Kodaira–Spencer map of the full universal group is a map of free R_0 -modules of rank $n - 1$. It is constructed on the formal deformation base by the Dieudonne and deformation formalism of [dJ95, Proposition 2.5.2, Definition 2.5.4 and Section 2.5.5]. Its reduction at the closed point is the universal tangent comparison, hence an isomorphism by [Ill85, Corollary 4.8(i) and (iii), pp. 179–180]. Its determinant is therefore a unit of R_0 , so the map is an isomorphism on R_0 and on every fibre after base change.

Ordinary and continuous differentials agree over R_0 . Indeed, perfection of k gives the finite expansion

$$f = \sum_{0 \leq \alpha_i < p} u^\alpha f_\alpha^p \quad (f \in R_0).$$

Thus every differential is a combination of the du_i , and the coordinate derivations prove their independence in both the ordinary and continuous differential modules.

Lemma 2.2. *Let S_0 be a finite etale frame stage over $B[1/x]$ containing the chosen level- m basis sections. Then:*

- (i) *The differentials $dy_{1,m}, \dots, dy_{d,m}$ form a basis of $\Omega_{S_0/k}$.*
- (ii) *The $y_{i,m}$ form a p -basis. For every $e \geq 1$, their monomials with each exponent less than p^e form a basis of S_0 over $S_0^{p^e}$.*

Proof. Let $Q = q^m$. Fix a geometric point $s : S_0 \rightarrow \Omega$ and a k -derivation $D : S_0 \rightarrow \Omega$ that kills $y_{1,m}, \dots, y_{d,m}$. Define

$$A_\epsilon = \Omega[\epsilon]/(\epsilon^2), \quad s_0(a) = s(a), \quad s_D(a) = s(a) + \epsilon D(a), \quad G_0 = s_0^* G, \quad G_D = s_D^* G.$$

Since $s_0(a)^Q = s_D(a)^Q$ for every $a \in S_0$, the base-change descriptions identify $G_0^{(Q)}$ and $G_D^{(Q)}$ with the same full p -divisible group G' . The factorization (2.1) gives isogenies

$$F_i : G_i \longrightarrow G', \quad V_i : G' \longrightarrow G_i, \quad V_i F_i = [p^m]_{G_i} \quad (i = 0, D).$$

Cancelling the fppf epimorphism F_i also gives $F_i V_i = [p^m]_{G'}$. Thus $C_i = \ker V_i$ is an embedded etale subgroup of $G'[p^m]$, with its chosen $(\mathbf{Z}_p/p^m)^d$ -basis.

For each basis section, its two coordinate values agree because $D(y_{j,m}) = 0$. The original coordinate generates the coordinate algebra of the full finite torsion group, including after base change. Consequently the sections agree in $G'[p^m]$. Their linear combinations use the same group law on G' , so the two embeddings of the constant group $(\mathbf{Z}_p/p^m)^d$ agree. Write $C = C_0 = C_D$. The two isogenies identify their targets with the fppf quotient:

$$(2.3) \quad \begin{array}{ccccc} G_0 & \xleftarrow[\sim]{\bar{V}_0} & G'/C & \xrightarrow[\sim]{\bar{V}_D} & G_D \\ & \searrow V_0 & \uparrow q_C & \nearrow V_D & \\ & & G' & & \end{array}$$

The induced isomorphism $\theta = \bar{V}_D \bar{V}_0^{-1} : G_0 \rightarrow G_D$ satisfies $\theta V_0 = V_D$. Modulo ϵ , both isogenies reduce to the same canonical V_m . Its fppf surjectivity therefore implies $\bar{\theta} = \text{id}_{s^* G}$.

Under the first-order crystalline comparison, the difference of the two Hodge filtrations is $\epsilon \text{KS}(D)$. The isomorphism θ , whose reduction is the identity, identifies these filtrations. Hence $\text{KS}(D) = 0$. The Kodaira–Spencer isomorphism of Subsection 2.2, base-changed to Ω , shows that D vanishes on the deformation base. On the algebraized crystal, the connection comes from the comparison along the PD diagonal [dJ95, Remark 2.2.4(d), pp. 19–20]; its first infinitesimal restriction is $1 + \epsilon \nabla_D$. Ordinary differentials are the free module described in Subsection 2.2. Etaleness of $S_0/B[1/x]$ then gives $D = 0$ on S_0 .

The module $\Omega_{S_0/k}$ is locally free of rank d . At every geometric point its dual has zero annihilator on $dy_{1,m}, \dots, dy_{d,m}$. These d differentials are therefore a basis, proving (i).

Relative Frobenius for the etale map $B[1/x] \rightarrow S_0$, together with the coordinate p -basis of B , makes S_0 free of rank p^d over S_0^p . Consider

$$(2.4) \quad S_0^p[Z_1, \dots, Z_d]/(Z_i^p - y_{i,m}^p) \longrightarrow S_0, \quad Z_i \longmapsto y_{i,m}.$$

Over a geometric point of S_0^p , the target is a local Artinian algebra of length p^d . The differences between the $y_{i,m}$ and their residues form a basis of its cotangent space. Nakayama’s lemma and nilpotence show that they generate its maximal ideal and that their p -block monomials span the algebra. There are p^d such monomials, so (2.4) is an isomorphism on every geometric fibre. Its source and target are free of the same finite rank over S_0^p , hence it is an isomorphism. Iterating its Frobenius twists proves (ii). These presentations persist under further etale extension, localization, and filtered colimits. $\square$

The independence in Lemma 2.2 is asserted on the ambient frame cover, before quotienting by P .

3. COMPATIBLE FINITE HONDA ACTIONS

3.1. A strict connected frame. Adjoin the finite Honda constants and a root c of $c^{q-1} = x$. For $g \in H$, write the marked comparison as

$$J_g : F \longrightarrow \rho_g F, \quad a_g = J'_g(0).$$

Comparison of the leading terms in the p -series gives $\rho_g(x) = xa_g^{1-q}$. Thus $\rho_g(c) = c/a_g$ defines a lift of the action of H and respects its cocycle. Set

$$(3.1) \quad \widehat{F}(X, Y) = cF(X/c, Y/c), \quad U(T) = cV(T/c^q), \quad K_g(T) = \rho_g(c)J_g(T/c).$$

Then K_g is strict and

$$[p]_{\widehat{F}}(T) = U(T^q), \quad U : \widehat{F}^{(q)} \longrightarrow \widehat{F}, \quad K_g : \widehat{F} \longrightarrow \rho_g \widehat{F}.$$

Choose a geometric isomorphism $\lambda_0 : \mathbb{H}_h \longrightarrow \widehat{F}$ with linear coefficient one. Its inverse $\psi(T) = T + \sum_{j>1} e_j T^j$ satisfies successive equations

$$(3.2) \quad e_j^q - e_j = b_j,$$

where b_j depends only on previous coefficients and coefficients of $[p]_{\widehat{F}}$. Indeed, compare degree qj in $\psi([p]_{\widehat{F}}(T)) = \psi(T)^q$. Each equation is monic with derivative -1 . The chosen coefficients consequently spread over finite etale components of the normal ambient base. The genuine homomorphism identities hold on those components, because they hold in the generic field and the components inject into that field.

Retain all components of the torsor of genuine connected frames. Its finite coefficient truncations have finite fibres, as quotients of the Honda automorphism group, and are unions of finite etale components. Composition with K_g preserves them. On the full frame torsor,

$$(3.3) \quad \rho_g(\lambda_0) = K_g \circ \lambda_0, \quad U \circ \lambda_0^{(q)} = \lambda_0.$$

The second identity follows from the p -series identity after substitution of T^q ; substitution by T^q is injective on formal series.

Let S be the joint ind-etale ambient algebra of the compatible etale frames, connected frames, and the constant and Kummer extensions of Subsection 3.1. Let

$$Q_m = q^m, \quad z_{i,m} = c^{Q_m} y_{i,m}, \quad F_m = \widehat{F}^{(Q_m)}, \quad \lambda_m = \lambda_0^{(Q_m)}.$$

Since $dz_{i,m} = c^{Q_m} dy_{i,m}$, Lemma 2.2 gives

$$(3.4) \quad S = S^{Q_m}[z_{1,m}, \dots, z_{d,m}]$$

with the Q_m -block monomials as a basis. All coefficients of F_m and λ_m belong to S^{Q_m} .

Proposition 3.1. *The following statements hold.*

(i) *There is a coaction of $\mathbb{H}_h[p^m]^d$ on S given by*

$$(3.5) \quad \alpha_m(z_{i,m}) = F_m(z_{i,m}, \lambda_m(T_i)), \quad \alpha_m(a) = a \quad (a \in S^{Q_m}).$$

(ii) *This coaction identifies the Frobenius diagonal with the Honda torsor:*

$$(3.6) \quad S \otimes_{S^{Q_m}} S \simeq S \otimes_k C_m, \quad C_m = k[T_1, \dots, T_d]/(T_i^{Q_m}),$$

where C_m carries the Honda group law.

Proof. Fix m and set $Q = Q_m$. Let A_m be the complete finite B -algebra of the original Weierstrass kernel of V_m , with coordinate Y . The coordinate Y is topologically nilpotent. Each chosen basis section gives a homomorphism $A_m \rightarrow S$, $Y \mapsto y_{i,m}$, after localization. For simultaneous sections, use the completed tensor product over B with one copy of A_m for each section; it is again finite and complete over B .

Continuous evaluation gives a ring homomorphism

$$B[[X, W_1, \dots, W_t]] \longrightarrow A_m[[W_1, \dots, W_t]], \quad a \longmapsto a^Q, \quad X \longmapsto Y, \quad W_j \longmapsto W_j.$$

It preserves all formal identities of the original group law. Write $F(X, Y) = \sum F_{a,b} X^a Y^b$. Equation (3.1) gives

$$F_m(c^Q Y, W) = c^Q F^{(Q)}(Y, W/c^Q).$$

For any fixed truncation in W , evaluation is performed in $A_m[[W]]$ first. Truncation then introduces finitely many powers of c^{-1} , and the resulting coefficients map to S .

For $z = c^Q y$, regrouping $a = r + Qj$, with $0 \leq r < Q$, makes the coefficient of $z^r W^b$ equal to

$$(3.7) \quad \left(c^{1-r-b} \sum_{j \geq 0} F_{r+Qj,b} y^j \right)^Q.$$

The inner series is evaluated in the complete algebra A_m . Thus, modulo W^N , the evaluated expression is a finite polynomial in z and W with coefficients in S^Q . Evaluation in the completed tensor product of the copies of A_m preserves formal identities involving several torsion coordinates.

Now work in $S[T_1, \dots, T_d]/(T_i^Q)$ and substitute the finite truncation of $\lambda_m(T_i)$ for W . Define

$$w_i = F_m(z_{i,m}, \lambda_m(T_i)).$$

Every element of the augmentation ideal has Q th power zero, so $w_i^Q = z_{i,m}^Q$. The block presentation (3.4) has the explicit algebra form

$$(3.8) \quad S \simeq S^Q[Z_1, \dots, Z_d]/(Z_i^Q - z_{i,m}^Q).$$

Its universal property supplies a unique S^Q -algebra homomorphism α_m sending $z_{i,m}$ to w_i .

For an increment δ with $\delta^Q = 0$, the regrouped evaluation at $z + \delta$ has the same coefficients (3.7), since $(z + \delta)^Q = z^Q$. The homomorphism α_m fixes those coefficients. Consequently, applying α_m to an evaluated expression agrees with finite substitution of $\alpha_m(z)$ for z . For associativity of α_m , evaluate the associativity identity of F in the complete torsion algebra with formal increment variables, truncate their nilpotent powers, and then localize. The homomorphism identity for the genuine frame λ_m therefore gives

$$(\alpha_m \otimes 1)\alpha_m = (1 \otimes \Delta)\alpha_m \quad \text{in } S[T_1, \dots, T_d, U_1, \dots, U_d]/(T_i^Q, U_i^Q),$$

where Δ is the Honda comultiplication. Equality holds on S^Q and on each generator of (3.8). Substitution $T_i = 0$ gives the counit, proving (i).

By (3.8), the source of the canonical torsor map is

$$S \otimes_{S^Q} S \simeq S[\epsilon_1, \dots, \epsilon_d]/(\epsilon_i^Q).$$

The map sends ϵ_i to $w_i - z_{i,m}$. The coefficient of T_i in this image is the value of the unit $\partial_2 F^{(Q)}(Y, 0)$ in the complete torsion algebra, multiplied by the linear coefficient of the strict frame. It is a unit before localization and remains a unit in S . Thus the induced map on augmentation ideals modulo their squares is invertible. Successive correction along the finite nilpotent filtration constructs an inverse polynomial. This proves (ii). $\square$

Proposition 3.2. *The actions in Proposition 3.1 have the following properties.*

(i) *Their restrictions satisfy*

$$(3.9) \quad \alpha_m|_{\mathbb{H}_h[p^\ell]^d} = \alpha_\ell \quad (\ell < m).$$

(ii) *They commute with the original action of H .*

(iii) *Under a change of connected and etale frames they satisfy*

$$(3.10) \quad (\kappa \otimes 1)\alpha_m = (1 \otimes \phi_\kappa^*)\alpha_m\kappa, \quad \phi_\kappa = [M]^{-1}b_{\text{diag}},$$

where $M \in \text{GL}_d(\mathbf{Z}_p)$ changes the ordered etale basis and $b \in \text{Aut}(\mathbb{H}_h)$ changes the connected frame.

Proof. Equations (2.2) and (3.3) give

$$(3.11) \quad z_{i,m-1} = U^{(q^{m-1})}(z_{i,m}), \quad U^{(q^{m-1})} \circ \lambda_m = \lambda_{m-1}.$$

Restrict α_m to level ℓ . Every augmentation element then has zero q^ℓ th power, so this restricted map fixes S^{q^ℓ} . The transition maps in (3.11) down to level ℓ have fixed coefficients. Their homomorphism identities show that the restricted coaction takes $z_{i,\ell}$ to $F_\ell(z_{i,\ell}, \lambda_\ell(T_i))$. It agrees with α_ℓ on the power subring and on its block basis. This proves (3.9). The transition evaluations are justified by formula (3.7), with modulus q^ℓ .

For $g \in H$, the comparison formulas are

$$\rho_g(z_{i,m}) = K_g^{(Q_m)}(z_{i,m}), \quad \rho_g(\lambda_m) = K_g^{(Q_m)} \circ \lambda_m.$$

All coefficients of $K_g^{(Q_m)}$ are fixed by α_m . Its homomorphism identity, evaluated by (3.7), therefore gives

$$(3.12) \quad (\rho_g \otimes 1)\alpha_m = \alpha_m \rho_g.$$

After fixing c and the finite constants, a deck transformation satisfies, in column notation,

$$\kappa(z_m) = [M]_{F_m} z_m, \quad \kappa(\lambda_m) = \lambda_m \circ b.$$

The coefficients of b lie in $\mathbf{F}_q$, since b commutes with T^q . Substitution in (3.5) proves (3.10) on each basis element, and both sides agree on the power subring. $\square$

Remark 3.3. The evaluated coefficient of the invariant differential in Proposition 3.1 can depend on z . For $p = 2$ and $h = 1$, the first Honda kernel has law $S + T + ST$ modulo (S^2, T^2) . A coaction $\alpha(f) = f + D(f)T$ satisfies $D^2 = D$. If $\alpha(z) = z + A(z)T$, then $D(A) = A$. Formula (3.7) fixes the formal coefficients in S^2 and retains the dependence of the evaluated coefficient $A(z)$ on z .

For $p = 2$ and $h = 2$, the first kernel has law $S + T + S^2T^2$ modulo (S^4, T^4) . Its invariant differential is dT , and the coefficient of S^2T^2 is one because $[2](T) = T^4$. Writing $\alpha(f) = \sum_{j=0}^3 D_j(f)T^j$, coassociativity gives

$$D_1^2 = 0, \quad D_2^2 = D_1, \quad D_1D_2 = D_2D_1 = D_3, \quad D_2^4 = 0.$$

The identity $D_2^2 = D_1$ records the term S^2T^2 . For $p = 2$, $h = 2$, and $m = 2$, equation (3.11) reads $U^{(4)}\lambda_2 = \lambda_1$.

4. A STABLE PRIME AND ITS RELATIVE ORBIT SUBGROUP

Assume that P satisfies the hypotheses of Theorem 2.1, and let

$$A = B/P, \quad K_0 = \text{Frac}(A), \quad a = \dim A.$$

Then $1 \leq a < d$. The lower bound follows because x is a nonzero element of the maximal ideal of A ; the upper bound follows because P is nonzero. In particular, no such prime exists when $d = 1$. We henceforth take $d \geq 2$ and work with an open subgroup of the norm-one group.

4.1. A valuation and the full frame component. We first arrange that H preserves a centered discrete valuation. Normalize the blowup of the maximal ideal of A . Excellence makes this normalization finite over the blowup. The finitely many prime divisors of the pullback of the maximal ideal define the centered Rees valuations of K_0 , and H permutes them. Choose finitely many elements of A distinguishing these valuations. For sufficiently large N , their values are smaller than every $Nv_i(\mathfrak{m}_A)$. An element of H fixing these elements modulo $\mathfrak{m}_A^N$ cannot permute the valuations. Continuity thus provides an open subgroup preserving each of them. Fix one of these valuations, denoted v . The action of H is isometric and extends to the completion of its valuation ring.

Lemma 4.1. *No open subgroup of H fixes A pointwise.*

Proof. If such a subgroup fixed A , its marked comparison isomorphisms would give an injective homomorphism into $\text{Aut}(F_{K_0})$. Indeed, equality of comparisons over K_0 is equality over the domain A , and reduction at the original closed point would identify their height- n markings. After choosing a geometric Honda frame of height h , the target embeds in $\mathcal{O}_{D_h}^\times$.

This homomorphism is continuous. Every finite tuple of Honda coefficients belongs to a finite set over $\mathbf{F}_q$. Conjugating comparison jets by fixed frame jets is a finite algebraic operation; continuity of the coefficients of the marked comparisons makes these finite tuples locally constant. Their truncations define the usual topology on $\mathcal{O}_{D_h}^\times$.

The Lie algebra of H is the simple $\mathbf{Q}_p$ -Lie algebra $\mathfrak{sl}_1(D_n)$ of dimension $n^2 - 1$. The target has dimension $h^2 < n^2 - 1$, so the differential of the homomorphism is zero. Continuous homomorphisms of p -adic analytic groups are analytic by [DDMS99, Theorem 9.4, p. 219]. Comparing the homomorphism with the trivial map, [DDMS99, Theorem 9.11(ii), p. 229] gives an open kernel. This contradicts injectivity. $\square$

We next select a component of the compatible connected and etale frame tower. The finite constant and Kummer algebra over K_0 has the lifted H -action $\rho_g(c) = c/a_g$. Its finitely many primitive idempotents are permuted locally constantly. After an open restriction, choose a factor field L_0 preserved by H . It contains c and the finite Honda constants.

Over L_0 , let D be the deck group of the remaining joint connected and etale frame torsor. It is an open subgroup of

$$\mathcal{O}_{D_h}^\times \times \text{GL}_d(\mathbf{Z}_p).$$

Choose a geometric frame and let $\Gamma \subset D$ be the Galois image over L_0 . The compatible component set is $\Gamma \backslash D$. The lifted action of H commutes with deck changes. Hence H acts on the component set through

$$(4.1) \quad \text{Aut}_D(\Gamma \backslash D) = N_D(\Gamma)/\Gamma.$$

The component action in (4.1) is continuous. At a fixed finite frame stage, choose a basis of its finite etale K_0 -algebra. Frame composition involves only finite jets. Torsion evaluation is also continuous: the comparison coefficients are integral, whereas the original torsion coordinate has positive valuation, so the tail has a uniform valuation bound. Passing to basis coordinates introduces only finitely many fixed denominators. The action on each primitive idempotent is therefore continuous. These idempotents form a finite discrete subset of a finite-dimensional valued vector space, so their permutation is locally constant. The inverse limit of these finite actions is the topology in (4.1). Applied to the finite constant and Kummer stage, continuity gives the open stabilizer used to select L_0 .

The group $N_D(\Gamma)/\Gamma$ is p -adic analytic by [DDMS99, Theorem 9.6(i) and (ii), p. 220], since $N_D(\Gamma)$ and Γ are closed. Its dimension is at most $h^2 + d^2$, whereas

$$n^2 - 1 = h^2 + d^2 + 2hd - 1 > h^2 + d^2.$$

Simplicity of $\mathfrak{sl}_1(D_n)$ forces the differential of $H \rightarrow N_D(\Gamma)/\Gamma$ to vanish. By [DDMS99, Theorem 9.11(ii), p. 229], the kernel of this homomorphism is open and fixes every component of the compatible tower.

Choose a prime above P in one component of the joint frame tower; the restricted action of H preserves it. Localize S at this prime. Denote the localized ring by R , its prime by P_R , and its residue field by K . Thus K/K_0 is separable algebraic, and

$$(4.2) \quad K/L_0 \text{ is Galois with group } \Gamma.$$

Indeed, a component of the full frame torsor is the field defined by the kernel of the Galois frame representation over the selected field L_0 .

The Honda coactions extend to R . The image of an element being inverted is its invertible augmentation plus a nilpotent element, and is therefore invertible. A coaction also fixes idempotents, by uniqueness of lifting idempotents across a nilpotent ideal. All identities of Propositions 3.1 and 3.2 consequently persist on R .

4.2. The relative imperfection degree. Let $\Lambda_0 = K_0^H$ and $\Lambda = K^H$, for the subgroup selected in Subsection 4.1.

Lemma 4.2. *The invariant fields satisfy the following properties.*

- (i) *The extensions K_0/Λ_0 and K/Λ are p -separable.*
- (ii) *The extension Λ/Λ_0 is separable algebraic.*
- (iii) *There is an isomorphism*

$$(4.3) \quad \Omega_{K/\Lambda} \simeq K \otimes_{K_0} \Omega_{K_0/\Lambda_0}.$$

- (iv) *For $r = \dim_K \Omega_{K/\Lambda}$, one has $1 \leq r \leq a < d$.*

Proof. For any field L with an action of H , let $\Lambda_L = L^H$. A p -basis of Λ_L remains p -independent in L . To see this, take a minimal L^p -linear dependence among Λ_L^p -independent monomials and normalize one coefficient to one. Subtracting any transform of this relation gives a shorter relation, so every normalized coefficient is invariant. But $(L^p)^H = (L^H)^p$, because a p th root in a field is unique. This contradicts the original independence. This proves the asserted p -separability.

The p -degree of K_0 is p^a . Indeed, complete local Noether normalization makes it finite over the fraction field of $k[[s_1, \dots, s_a]]$, and finite field extensions preserve finite p -degree. Separable algebraic extensions and their filtered unions preserve a p -basis, so $[K : K^p] = p^a$ as well.

Suppose that $\Omega_{K_0/\Lambda_0} = 0$. Then there is a p -basis $f_1, \dots, f_a$ of K_0 contained in Λ_0 . For the original local generators t_i of A , write

$$(4.4) \quad t_i = \sum_{0 \leq \alpha_j < p} c_{i,\alpha}^p f^\alpha.$$

Let $\delta_g = \min_i v(gt_i - t_i)$, allowing the value $+\infty$. For any fixed power series in the t_i over k , its displacement has valuation at least δ_g : monomial differences telescope, and centeredness of the valuation makes the resulting sums converge. The quotient formula therefore gives, for a fixed fraction $c \in K_0$,

$$v(gc - c) \geq \delta_g - C_c$$

for g in a sufficiently small open subgroup of H . There are only finitely many coefficients $c_{i,\alpha}$. Applying g to (4.4), and using the invariance of the f_j , yields a constant C' with

$$\delta_g \geq p\delta_g - C'.$$

Every finite δ_g is thus at most $C'/(p-1)$. Continuity provides a neighborhood of the identity on which all δ_g exceed this bound; they are then infinite. An open subgroup fixes A pointwise, contrary to Lemma 4.1. Hence $\Omega_{K_0/\Lambda_0} \neq 0$.

If $e \in \Lambda$, its minimal polynomial over K_0 is fixed by H , so its coefficients belong to Λ_0 . Thus Λ/Λ_0 is algebraic. It is separable, since K_0/Λ_0 is p -separable and K/K_0 is separable algebraic. The exact sequences of differentials for Λ/Λ_0 and K/K_0 give (4.3). The positive rank is at most the absolute p -rank a , proving the inequalities. $\square$

4.3. The relative orbit algebra. Write $\text{ev} : R \rightarrow K$ for reduction and let

$$\beta_m = (\text{ev} \otimes 1)\alpha_m : R \rightarrow K \otimes_k C_m.$$

Let $Q = Q_m$. Localizing the canonical torsor map gives

$$R \otimes_{R^Q} R \simeq R \otimes_k C_m, \quad r \otimes s \mapsto r\alpha_m(s).$$

Each inverted element has invertible augmentation under the coaction, and its difference from that augmentation is nilpotent. Evaluating the first factor by ev and quotienting the second by P_R gives

$$\frac{K \otimes_k C_m}{(\beta_m(P_R))} \simeq K \otimes_{R^Q} (R/P_R) = K \otimes_{R^Q} K \simeq K \otimes_{K^Q} K.$$

The last isomorphism uses the surjectivity of $\text{ev} : R \rightarrow K$, which gives $\text{ev}(R^Q) = K^Q$. Every $\beta_m(b)$ with $\text{ev}(b) \neq 0$ is already a unit.

For every $e \in \Lambda$ and every $a, b \in R$ with $\text{ev}(b) \neq 0$ and $e = \text{ev}(a)/\text{ev}(b)$, impose

$$(4.5) \quad \beta_m(a) - e\beta_m(b) = 0.$$

In $K \otimes_{K^Q} K$, each numerator is the unit $1 \otimes \text{ev}(b)$ times $1 \otimes e - e \otimes 1$. The relations for all $e \in \Lambda$ therefore impose the balancing over $K^Q\Lambda$. Thus the ideal I_m generated by the elements $\beta_m(P_R)$ and all the numerators in (4.5) has quotient

$$(4.6) \quad D'_m = (K \otimes_k C_m)/I_m = K \otimes_{K^Q\Lambda} K.$$

Commutation with H in (3.12), stability of P_R , and invariance of e imply that I_m is stable under coefficientwise action of H . Its unique reduced row echelon basis in the constant monomial basis has invariant coefficients. Therefore

$$(4.7) \quad I_m = K \otimes_\Lambda J_m$$

for an ideal $J_m \subset \Lambda \otimes_k C_m$. Closure under multiplication descends by faithful flatness.

Lemma 4.3. *The ideal J_m is a Hopf ideal. Thus $Y_m = \text{Spec}((\Lambda \otimes_k C_m)/J_m)$ is a finite subgroup scheme of $\mathbb{H}_h[p^m]_\Lambda^d$.*

Proof. Let $\mathcal{B}_m = \Lambda \otimes_k C_m$, let $A_m = \mathcal{B}_m/J_m$, and write $\pi_m : \mathcal{B}_m \rightarrow A_m$ for the quotient map. By (4.6) and (4.7), $D'_m = K \otimes_\Lambda A_m$. The second-factor inclusion $K \rightarrow K \otimes_{K^Q\Lambda} K$ and the quotient map satisfy

$$(4.8) \quad \begin{aligned} \eta_m : K &\rightarrow K \otimes_\Lambda A_m, & \nu_m &= 1_K \otimes \pi_m, \\ \nu_m \beta_m &= \eta_m \text{ev}, & \eta_m(e) &= e \otimes 1 \quad (e \in \Lambda). \end{aligned}$$

The last equality uses the relations for every invariant fraction.

Suppress the subscript m temporarily and let $\Delta_{\mathcal{B}}$ be the coproduct on $\mathcal{B} = \Lambda \otimes_k C_m$. There are two Λ -algebra maps with common target $K \otimes_\Lambda A \otimes_\Lambda A$:

$$\delta = 1_K \otimes ((\pi \otimes \pi)\Delta_{\mathcal{B}}), \quad \theta = \eta \otimes \pi.$$

The map δ is K -linear, whereas θ acts on K through η . For $j \in J$ one has $\theta(1 \otimes j) = 1 \otimes 1 \otimes \pi(j) = 0$. Since $I = K \otimes_\Lambda J$, one has $\theta(I) = 0$. On evaluated ambient elements, (4.8) and coassociativity give the exact identity

$$(4.9) \quad \begin{aligned} \theta\beta &= ((\nu\beta) \otimes \pi)\alpha \\ &= (\text{ev} \otimes \pi \otimes \pi)(\alpha \otimes 1)\alpha \\ &= (\text{ev} \otimes \pi \otimes \pi)(1 \otimes \Delta)\alpha = \delta\beta. \end{aligned}$$

Here π on C_m denotes the map $c \mapsto \pi(1 \otimes c)$. Equivalently, the following square commutes:

$$\begin{array}{ccc} R & \xrightarrow{\alpha} & R \otimes_k C_m \\ \beta \downarrow & & \downarrow (\nu\beta) \otimes \pi \\ K \otimes_\Lambda \mathcal{B} & \xrightarrow{\theta} & K \otimes_\Lambda A \otimes_\Lambda A. \end{array}$$

For $a \in P_R$, equation (4.9) implies $\delta\beta(a) = \theta\beta(a) = 0$. For a generator $z = \beta(a) - e\beta(b)$ associated with an invariant fraction, the equality $\eta(e) = e \otimes 1$ implies

$$\delta(z) = \theta\beta(a) - e\theta\beta(b) = \theta(z) = 0.$$

These are all the generators of I , so $\delta(I) = 0$. Faithful flatness of K/Λ and exactness of tensor products over Λ therefore give

$$(4.10) \quad (\pi \otimes \pi)\Delta_{\mathcal{B}}(J) = 0, \quad \Delta_{\mathcal{B}}(J) \subset J \otimes_\Lambda \mathcal{B} + \mathcal{B} \otimes_\Lambda J.$$

The counit vanishes on both kinds of generator, since their values are $\text{ev}(a)$ for $a \in P_R$ and $\text{ev}(a) - e\text{ev}(b)$, respectively. It therefore vanishes on J . Consequently $\text{Spec } A$ is a closed submonoid of the ambient Honda group. That group is commutative and killed by p^m , so its inversion morphism is multiplication by $p^m - 1$. Repeated multiplication restricts to $\text{Spec } A$; hence the antipode also preserves J . $\square$

Proposition 4.4. *There is a formal subgroup $Y \subset (\mathbb{H}_h^d)_\Lambda$ satisfying:*

- (i) *The group Y is smooth and p -divisible, with dimension r and height hr .*
- (ii) *For every $m \geq 1$, one has the scheme-theoretic equality*

$$Y[p^m] = Y \cap \mathbb{H}_h[p^m]^d = Y_m.$$

Proof. Restriction $C_{m+1} \rightarrow C_m$ takes each equation of P_R and each numerator in (4.5) to its equation at level m , using the same R , ev , P_R , K , Λ , and representatives a, b . Conversely, every generator at level m is such an image. Compatibility (3.9) therefore gives

$$(4.11) \quad \text{im}(I_{m+1}) = I_m, \quad \text{im}(J_{m+1}) = J_m,$$

where the second equality follows from (4.7) and faithful flatness over Λ . Let $\mathcal{B}_\infty = \Lambda[[T_1, \dots, T_d]]$ and $\mathfrak{b}_m = (T_1^{q^m}, \dots, T_d^{q^m})$. These ideals are cofinal with powers of the augmentation ideal. Surjectivity in (4.11) lifts each element of J_m successively through every higher level. The inverse limit is therefore a closed ideal $J \subset \mathcal{B}_\infty$ with exact quotients

$$(4.12) \quad \mathcal{B}_\infty/(J + \mathfrak{b}_m) = (\Lambda \otimes_k C_m)/J_m = A_m.$$

For $f \in J$, the formal coproduct $\Delta(f)$ has zero image modulo

$$J(S) + J(T) + (S_i^{q^m}, T_i^{q^m})$$

for every m , by (4.10). The ring $\Lambda[[S, T]]/(J(S) + J(T))$ is Noetherian and complete, and these truncation ideals are cofinal with its maximal-ideal powers. Separatedness therefore gives the formal coproduct containment. The counit and antipode identities pass to the limit by the same separatedness. Consequently

$$Y = \text{Spf } C, \quad C = \Lambda[[T_1, \dots, T_d]]/J$$

is a formal subgroup, and (4.12) identifies its scheme-theoretic intersection with each finite kernel as Y_m .

By Lemma 4.2, a p -basis $b = (b_1, \dots, b_s)$ of Λ extends to a p -basis $(b, t_1, \dots, t_r)$ of K . For $Q = Q_m$ this gives

$$K = K^Q[b, t_1, \dots, t_r], \quad \Lambda = \Lambda^Q[b], \quad K^Q \Lambda = K^Q[b].$$

The monomials $t_1^{i_1} \cdots t_r^{i_r}$ with $0 \leq i_j < Q$ form a basis of K over $K^Q \Lambda$. Hence

$$(4.13) \quad \begin{aligned} D'_m &\simeq K[X_1, \dots, X_r]/(X_i^Q - t_i^Q) \\ &\simeq K[\epsilon_1, \dots, \epsilon_r]/(\epsilon_i^Q), \quad \epsilon_i = X_i - t_i. \end{aligned}$$

Its augmentation is multiplication of the two factors of K . Faithful extension from Λ to K shows that $A_m = \mathcal{O}(Y_m)$ has length Q_m^r and cotangent dimension r . Let $\mathfrak{n}$ be the augmentation ideal of C and let $\mathfrak{a}_m$ be generated by the images of $T_i^{Q_m}$. Then

$$\mathfrak{n}^{d(Q_m-1)+1} \subset \mathfrak{a}_m \subset \mathfrak{n}^{Q_m}.$$

The lengths $\ell(C/\mathfrak{a}_m) = Q_m^r$ and Hilbert–Samuel growth imply $\dim C = r$. Since $\mathfrak{a}_1 \subset \mathfrak{n}^2$, the cotangent dimension of C is also r . Its coefficient field is the given field Λ , so lifts of a cotangent basis give a surjection $\Lambda[[w_1, \dots, w_r]] \rightarrow C$. A nonzero kernel would lower the dimension. This is an isomorphism, and C is geometrically regular.

The inherited p -map induces a Λ -algebra endomorphism $f : C \rightarrow C$ with $f(T_i) = T_i^q$ and $f(\mathfrak{n})C = \mathfrak{a}_1$. Choose lifts in C of a Λ -basis of $C/\mathfrak{a}_1$. Successive approximation in the $\mathfrak{a}_1$ -adic topology writes each element of C as a linear combination of these lifts with coefficients in $f(C)$: at the N th step the error belongs to $f(\mathfrak{n})^N C$, and the correction coefficients belong to $f(\mathfrak{n}^N)$. Completeness permits passage to the limit. Hence C is finite as a module over C through f .

Finite maps preserve dimension, so $\dim(C/\ker f) = \dim C = r$. Since C is a power-series domain, $\ker f = 0$. The images under f of a system of parameters of C form a system of parameters in the regular target ring and hence a regular sequence. This finite module has depth r over the regular local domain C ; the Auslander–Buchsbaum formula makes it free. Its rank is $\dim_\Lambda(C/\mathfrak{a}_1) = q^r$. Therefore $[p]$ is finite flat, and iteration gives finite flat degree q^{mr} for $[p^m]$. Its coordinate substitution is $T_i \mapsto T_i^{q^m}$, whose kernel scheme has coordinate ring $C/\mathfrak{a}_m = A_m$ by (4.12). Thus $Y[p^m] = Y_m$, and Y is p -divisible of height hr . $\square$

4.4. Constant equations and subgroup descent. We show that Y is defined over $\mathbf{F}_q$. Let $\Lambda' = \Lambda^{\text{perf}}$, let $\mathcal{B}' = \Lambda'[[T_1, \dots, T_d]]$, and let $J' = \overline{J\mathcal{B}'}$ be the closed extended ideal. For the augmentation ideal $\mathfrak{n}$ of C , finite jets give

$$\mathcal{B}'/J' \simeq \varprojlim_N ((C/\mathfrak{n}^N) \otimes_\Lambda \Lambda') = C \widehat{\otimes}_\Lambda \Lambda' \simeq \Lambda'[[w_1, \dots, w_r]].$$

The displayed inverse limit defines the completed tensor product. The last isomorphism follows from Proposition 4.4, so the quotient is a domain and is reduced. Let σ act on coefficients by q -Frobenius. Stability under the p -map gives, for $f \in J'$,

$$f(T^q) = (\sigma^{-1}f(T))^q \in J'.$$

Reducedness implies $\sigma^{-1}J' \subset J'$. Since σ preserves height, this inclusion of prime ideals is an equality. The unique reduced row echelon basis of each finite jet of J' is fixed by σ . Its coefficients therefore satisfy $a^q = a$. They lie in $\mathbf{F}_q$, which was already included among the finite flags. The compatible jets define a single formal subgroup over $\mathbf{F}_q$. Equality with the defining ideal of Y is checked on every finite jet, where scalar extension from Λ to Λ' is faithfully flat. Closedness then gives equality of the formal ideals over Λ .

Let κ be a deck transformation preserving the selected prime. It commutes with the action of H and therefore preserves Λ . Formula (3.10) transports the equations of P_R and every equation (4.5). The action of κ on the coefficients in $\mathbf{F}_q$ is trivial, so ϕ_κ preserves Y .

Choose an open normal subgroup $\Gamma_1 \subset \Gamma$, and let $K_1 = K^{\Gamma_1}$. By (4.2), this field is finite over L_0 , and therefore over K_0 . It is preserved by H , because the actions of H and Γ commute. The finite flags are already in L_0 and are fixed by the deck group.

Let $T = T_p G_e$ and

$$(4.14) \quad U = T^\vee \otimes_{\mathbf{Z}_p} G_c = \text{Hom}_{\mathbf{Z}_p}(T, G_c).$$

Here $\text{Hom}_{\mathbf{Z}_p}(T, G_c)$ denotes the twist of G_c by the dual Tate local system $T^\vee$. At level m it is finite etale descent of d copies of $G_c[p^m]$, and the compatible Tate transitions make it a p -divisible group.

Over K , an ordered Tate basis $a = (a_1, \dots, a_d)^t$ and the connected frame λ identify $\mathbb{H}_h^d$ with U : the tuple u corresponds to the homomorphism $a_i \mapsto \lambda(u_i)$. If $\kappa(a) = Ma$ and $\kappa(\lambda) = \lambda b$, the transformed coordinates satisfy $Mu' = b_{\text{diag}}u$, or $u' = \phi_\kappa u$. Thus (3.10) is precisely the descent cocycle of (4.14). Each Y_m descends to a closed finite flat subgroup of $U[p^m]$ over K_1 . Finite presentation reduces the descent at each level to ordinary finite Galois descent. Uniqueness of Galois descent preserves the transitions of Proposition 3.2 and the kernel identities of Proposition 4.4. We obtain

$$(4.15) \quad Y_1 \longrightarrow U, \quad \dim Y_1 = r, \quad \text{ht } Y_1 = hr, \quad 0 < r < d.$$

For the comparison convention $j_g : G \longrightarrow g^*G$, let c_g, t_g be the connected and Tate comparisons. The induced comparison on U is

$$f \longmapsto c_g \circ f \circ t_g^{-1}.$$

On the joint frame it acts coefficientwise and fixes the Honda parameters. The equations of Y lie in $\mathbf{F}_q$, fixed by H , so this comparison carries Y_1 to g^*Y_1 . The inclusion square and the cocycle for H descend, proving covariance of the subgroup inclusion (4.15).

5. THE VALUED REALIZATION AND THE VERTICAL PRIME THEOREM

Extend the centered valuation of K_0 to the finite extension K_1 . There are finitely many extensions. Continuity at a finite frame stage containing K_1 makes their permutation locally constant, so an open subgroup preserves one extension. Complete its valuation ring. The original full group over A extends by base change to this complete discrete valuation ring, with its original marked Honda special fibre. Its generic field still has $x \neq 0$. The subgroup in (4.15) and all comparison maps extend by field extension.

5.1. An invariant parameter. Let A_v be the completion of the chosen valuation ring of K_1 . The proof of Lemma 6.2 uses only valuation rings and Borger's residual-perfection construction. Apply Lemma 6.2 to A_v and its action of H . In the notation of its proof, let s be the original uniformizer, let u be the added fixed residue variable, and let u_0 be its canonical coefficient lift. The coefficient expansion in Lemma 6.2 is

$$(5.1) \quad u = u_0 + u_1 s + u_2 s^2 + \dots, \quad u_1 \in (k')^\times.$$

The lemma proves that $\pi = u - u_0$ has valuation one and is fixed by H . The extended valuation ring is $k'[[\pi]]$, where k' is perfect and H acts coefficientwise. The induced action on $W(k')[[\pi]]$ commutes with Witt Frobenius and with the lift $\varphi(\pi) = \pi^p$; it preserves the good and overconvergent coefficient bounds. Equivariant base change extends the full group over $k'[[\pi]]$, its generic subgroup, and all comparison maps.

5.2. The full crystalline module. Let M be the good rational contravariant Dieudonne module of G over $k'[[\pi]]$. In its Honda marking the special Frobenius is $A_0\sigma$, with A_0 rational, and $F_0^n = p\sigma^n$. For $g \in H$, the comparison on the special fibre is the original Honda matrix $R(g)$ followed by the coefficient automorphism induced by g . The group H fixes the finite Honda constants.

Let $\mathcal{E}$ be the generic Amice field and let $M_{\mathcal{E}} = M \otimes \mathcal{E}$. Over $\mathcal{E}$, let

$$M_{\text{et}} = \mathbb{D}(G_e), \quad C = \mathbb{D}(G_c), \quad Q = \mathbb{D}(Y_1).$$

The exact contravariant crystalline functor, applied to the connected–etale sequence and the inclusion (4.15), gives

$$(5.2) \quad 0 \longrightarrow M_{\text{et}} \longrightarrow M_{\mathcal{E}} \longrightarrow C \longrightarrow 0, \quad \mathbb{D}(U) \longrightarrow Q \longrightarrow 0.$$

These maps are horizontal and Frobenius compatible by crystalline functoriality and base change [dJ95, Remark 2.2.4(h), Sections 2.3.3–2.3.4 and the beginning of Section 4.3]. On an etale trivialization of T ,

$$(5.3) \quad M_{\text{et}} = T^{\vee} \otimes \mathbf{1}, \quad \mathbb{D}(U) = T \otimes C = \text{Hom}(M_{\text{et}}, C).$$

Contravariance on the transition matrices descends (5.3) with its connection, Frobenius, and comparisons for H . The subgroup inclusion square of Subsection 4.4 therefore gives the quotient $\text{Hom}(M_{\text{et}}, C) \longrightarrow Q$ with its original comparisons for H .

The etale group G_e has height d , the connected group becomes $\mathbb{H}_h$ geometrically, and Y_1 becomes the subgroup Y of Proposition 4.4. Thus $(\text{rank } M_{\text{et}}, \text{rank } M, \text{rank } C, \text{rank } Q) = (d, n, h, hr)$, with slope zero for M_{et} and $1/h$ for C, Q . Field extensions preserve $0 < r < d$. This proves Theorem 2.1.

Theorem 5.1. *Let $1 \leq h < n$. A nonzero prime of $k[[u_h, \dots, u_{n-1}]]$ avoiding u_h is stable under no open subgroup of either the height- n Morava stabilizer or its norm-one subgroup.*

Proof. If $d = 1$, the assertion follows because a one-variable power-series ring has no nonzero prime avoiding its parameter. If $d \geq 2$, suppose P is such a prime. Restrict to an open subgroup of the norm-one group contained in the prescribed group. All the subsequent restrictions in Theorem 2.1 are open in the norm-one group, and there are only finitely many of them. The prime remains stable.

Apply Theorem 2.1 to P . The resulting marked group and covariant subgroup Y_1 , with $0 < r < d$, satisfy the hypotheses of Theorem 6.1: take $\beta_g : g^*G \rightarrow G$ to be the inverse of $j_g : G \rightarrow g^*G$. Inverting the comparisons gives the required subgroup covariance squares. Theorem 6.1 gives a contradiction. $\square$

For $n = 1$ the range $1 \leq h < n$ is empty.

Corollary 5.2. *For every $0 \neq f \in B$ and every prescribed open subgroup H of the height- n Morava stabilizer or its norm-one subgroup, there are integers $N \geq 0$ and $s \geq 1$, elements $g_1, \dots, g_s \in H$, and coefficients $b_1, \dots, b_s \in B$ such that*

$$x^N = \sum_{i=1}^s b_i \rho_{g_i}(f).$$

Proof. Let I be the ideal generated by all $\rho_g(f)$, $g \in H$. If $I[1/x]$ were proper, its finitely many minimal primes, contracted to B , would be nonzero primes avoiding x , permuted by H . Their stabilizers are closed: ideals in the complete Noetherian local ring B are closed in the maximal-ideal topology, and the action of H is continuous. A closed finite-index subgroup of a compact group is open. Such a stabilizer contradicts Theorem 5.1. Therefore $I[1/x] = B[1/x]$. One expression for the unit involves only finitely many orbit elements. Clearing its denominators gives the asserted identity in B , with every g_i in H . $\square$

6. A CANONICAL TENSOR EXCLUSION THEOREM

6.1. Statement and equivariant coefficient rings. Let D_n be the endomorphism division algebra of the height- n Honda group, and let H be an open subgroup of $\text{SL}_1(D_n)(\mathbf{Q}_p)$. For an action of H on a ring A , comparisons on a p -divisible group G/A are isomorphisms $\beta_g : g^*G \longrightarrow G$ satisfying the cocycle identity. All Dieudonné modules in this section are contravariant. The notation $\mathbf{Q}_{p^n}$ means $W(\mathbb{F}_{p^n})[1/p]$.

Theorem 6.1. *Let A be a complete discrete valuation ring of characteristic p , with fraction field L . Suppose that a one-dimensional p -divisible group G/A of height n has the following properties.*

- (i) *The group H acts on A by automorphisms preserving the valuation, and G has comparisons for this action.*
- (ii) *Its special fibre is the marked height- n Honda group. The action fixes $\mathbb{F}_{p^n}$. After perfect residue extension its rational Dieudonné module has a marked basis in which*

$$F_0 = A_0\sigma, \quad A_0 \in \mathrm{GL}_n(\mathbf{Q}_p), \quad F_0^n = p\sigma^n.$$

The comparison on the special fibre is $R(g)\gamma_g$, where $R(g)$ is the original Honda matrix and γ_g is the induced coefficient action.

- (iii) *Over L , the connected part G_c has height h and the étale quotient G_e has height $d = n - h$, where $1 \leq h < n$.*

Let $T = T_p(G_e)$. There is no closed p -divisible subgroup

$$Y \subset U := T^\vee \otimes_{\mathbf{Z}_p} G_c$$

of dimension r and height hr , with $0 < r < d$, that is preserved by the comparisons induced on U .

The twist $U = \mathrm{Hom}_{\mathbf{Z}_p}(T, G_c)$ is obtained by descending G_c^d with the inverse changes of Tate basis. The hypotheses on Y in Theorem 6.1 concern its inclusion and comparisons over L .

Lemma 6.2. *Let A be a complete discrete valuation ring of characteristic $p > 0$, and let a group H act on A by automorphisms preserving its valuation. There is an H -equivariant extension of complete discrete valuation rings $A \rightarrow B$ of ramification index one with the following properties.*

- (i) *The ring B has the form $k[[t]]$, where k is perfect and every element of H fixes t . The action is coefficientwise in this presentation.*
- (ii) *Base change extends equivariant p -divisible groups over A , their comparisons and markings, and equivariant subgroup inclusions over $\mathrm{Frac}(A)$ to the corresponding objects over B and $\mathrm{Frac}(B)$.*

Proof. Choose a uniformizer $s \in A$, let κ be its residue field, and adjoin an indeterminate u fixed by H . The ring

$$A' = \widehat{A[u]_{(s)}}$$

is a complete discrete valuation ring with uniformizer s and residue field $\kappa(\bar{u})$. Each element of H preserves the prime (s) , so the action extends to A' and fixes u .

Choose a p -basis $\mathcal{P}$ of κ and a set $\mathcal{T} \subset A'$ lifting the p -basis $\mathcal{P} \cup \{\bar{u}\}$ of $\kappa(\bar{u})$, with $u \in \mathcal{T}$. Define

$$S = \kappa(\bar{u})[X_{b,j} : b \in \mathcal{T}, j \geq 1], \quad k = \mathrm{Frac}(S^{\mathrm{perf}}).$$

The coefficient construction of [Bor04, Section 1.9, Construction 1.11 and Theorem 1.14] identifies the generic residual perfection $B = (A')^g$ of [Bor04, Section 1.15] with $k[[s]]$. Let $c : k \rightarrow B$ be the coefficient section. The structural map $A' \rightarrow B$ satisfies

$$(6.1) \quad b = c(\bar{b}) + \sum_{j \geq 1} c(X_{b,j})s^j \quad (b \in \mathcal{T}).$$

The maps $S \rightarrow S^{\mathrm{perf}} \rightarrow k$ are injective because S is a domain. In particular $X_{u,1} \neq 0$, so $c(X_{u,1})$ is a unit. Equation (6.1) gives

$$(6.2) \quad v_B(u - c(\bar{u})) = 1.$$

Each automorphism $g : A' \rightarrow A'$ has ramification index one and induces an isomorphism of residue fields. [Bor04, Proposition 2.1] therefore gives a canonical lift $\tilde{g} : B \rightarrow B$ satisfying $\tilde{g} \circ \iota = \iota \circ g$, where $\iota : A' \rightarrow B$ is the structural map. Uniqueness gives

$$\widetilde{gh} = \tilde{g}\tilde{h}, \quad \widetilde{g^{-1}} = (\tilde{g})^{-1}.$$

The coefficient section is natural under these automorphisms, as in [Bor04, Section 1.2, equation (1.2.1)]. Explicitly, for $\alpha \in k$ choose a lift $b_m \in B$ of $\alpha^{p^{-m}}$ and write

$$c(\alpha) = \lim_{m \rightarrow \infty} b_m^{p^m}.$$

Consecutive terms differ by an element of $\mathfrak{m}_B^{p^m}$, and changing a lift has the same effect. Completeness gives a limit independent of the lifts. Characteristic p makes c additive and multiplicative. Applying $\tilde{g}$ to the limit gives

$$\tilde{g}(c(\alpha)) = c(\tilde{g}(\alpha)).$$

Since u and $\bar{u}$ are fixed by H , the element $t = u - c(\bar{u})$ is fixed as well. By (6.2), it is a uniformizer. Its expansion in s has invertible linear coefficient, so formal substitution has an inverse and $B = c(k)[[t]]$. Naturality of c and the equality $\tilde{g}(t) = t$ make the action coefficientwise. This proves (i).

Equivariant base change over A and $\text{Frac}(A)$ preserves the comparison maps, their cocycle identities, subgroup inclusions and special-fibre markings, proving (ii). $\square$

Apply Lemma 6.2 to the data of Theorem 6.1. Dimensions, heights and properness of Y are unchanged. Write

$$\mathcal{O} = W(k), \quad K = \mathcal{O}[1/p], \quad R = \mathcal{O}[[t]][1/p].$$

Let σ be Witt Frobenius and let ϕ be defined by $\phi|_{\mathcal{O}} = \sigma$ and $\phi(t) = t^p$. The coefficient action of each element of H commutes with ϕ and fixes dt . We also use the Amice field

$$\mathcal{E} = \left\{ \sum_{i \in \mathbb{Z}} a_i t^i : a_i \in K, \inf_i v_p(a_i) > -\infty, v_p(a_i) \rightarrow \infty \text{ as } i \rightarrow -\infty \right\}$$

and its bounded overconvergent subfield

$$\mathcal{E}^\dagger = \left\{ \sum_i a_i t^i \in \mathcal{E} : \liminf_{i \rightarrow -\infty} \frac{v_p(a_i)}{-i} > 0 \right\}.$$

All three rings have the connection defined by d/dt . Each element of H preserves their defining bounds, because its action preserves the valuation of every coefficient and every exponent of t . The residue field of the integral Amice ring is $k((t))$, with p -basis $\{t\}$.

6.2. The quotient furnished by the actual subgroup. Let M be the evaluation of the rational contravariant Dieudonné crystal of G on $\mathcal{O}[[t]]$. It is a finite free (ϕ, ∇) -module over R . By crystalline base change, [dJ95, Remark 2.2.4(h) and Sections 2.3.3–2.3.4], $M_{\mathcal{E}} := M \otimes_R \mathcal{E}$ is the module of the generic fibre of G .

Proposition 6.3. *The connected-étale sequence and the inclusion $Y \hookrightarrow U$ give modules M_{et}, C, Q with the following properties.*

- (i) *There are exact sequences of horizontal maps commuting with Frobenius and compatible with the comparisons for H :*

$$(6.3) \quad 0 \longrightarrow M_{\text{et}} \longrightarrow M_{\mathcal{E}} \longrightarrow C \longrightarrow 0,$$

$$(6.4) \quad q : V := \text{Hom}(M_{\text{et}}, C) \longrightarrow Q \longrightarrow 0.$$

- (ii) *The ranks of M_{et}, C, Q are d, h, hr , respectively, and their slopes are $0, 1/h, 1/h$.*

Proof. Apply the contravariant crystal functor to the connected-étale sequence and to $Y \hookrightarrow U$. Exactness can be checked on truncated Barsotti–Tate sequences before passing to the rational module; see [dJ95, Section 4.3, p. 41]. Thus $M_{\text{et}} = \mathbb{D}(G_e)$, $C = \mathbb{D}(G_c)$ and $Q = \mathbb{D}(Y)$.

On an étale trivialization of the Tate local system, contravariance and additivity give

$$\mathbb{D}(G_e) = T^\vee \otimes \mathbf{1}, \quad \mathbb{D}(U) = T \otimes C = M_{\text{et}}^\vee \otimes C.$$

These identities commute with the inverse transpose changes of Tate basis and descend to an identification $\mathbb{D}(U) = V$ of crystals with connection.

The étale module has slope zero. A one-dimensional connected group of height h is geometrically isoclinic of slope $1/h$. Consequently V , and hence its quotient Q , are isoclinic of that slope. Their ranks are their group heights.

Let $\beta_{U,g} : g^*U \rightarrow U$ and $\beta_{Y,g} : g^*Y \rightarrow Y$ be the given comparisons, and denote the inverses of their contravariant crystal maps by $\theta_{U,g}$ and $\theta_{Y,g}$. Applying the functor to the actual commuting square of group maps gives

$$q\theta_{U,g} = \theta_{Y,g} g^*q.$$

□

Let $s = hr$ and introduce the rational tensor functor

$$\mathcal{T}(N) = \bigwedge^s \text{End}(N)^\vee.$$

Restriction to M_{et} and projection to C give a surjection $\text{End}(M_{\mathcal{E}}) \rightarrow V$. Composing with q and dualizing its determinant gives a decomposable line preserved by H :

$$(6.5) \quad \ell = (\det Q)^\vee \subset \mathcal{T}(M_{\mathcal{E}}).$$

Scalar matrices act trivially on $\mathcal{T}$.

Lemma 6.4. (i) *The canonical subobject of minimum slope in $\mathcal{T}(M_{\mathcal{E}})$ is*

$$\mathcal{T}_{\min} = \bigwedge^s V^\vee = \bigwedge^s \text{Hom}(C, M_{\text{et}}),$$

of slope $-r$.

(ii) *The line ℓ is contained in $\mathcal{T}_{\min}$.*

Proof. The slopes of $\text{End}(M_{\mathcal{E}})^\vee$ are $-1/h, 0, 1/h$, with multiplicities $hd, d^2 + h^2, hd$. Its minimum subobject is $\text{Hom}(C, M_{\text{et}}) = V^\vee$, embedded using the quotient to C and the inclusion of M_{et} . Since $0 < s < hd$, the minimum possible sum of s slopes is $-s/h = -r$ and occurs precisely on $\bigwedge^s V^\vee$. Taking determinants of $Q^\vee \subset V^\vee$ gives $\ell \subset \mathcal{T}_{\min}$. □

6.3. A minimal overconvergent hull. Over $\mathcal{E}^\dagger$ or $\mathcal{E}$, a subobject is a finite-dimensional submodule whose Frobenius linearization is invertible and whose connection is induced from the ambient module. Let $\mathcal{T}(M)^\dagger = \mathcal{T}(M) \otimes_R \mathcal{E}^\dagger$.

Proposition 6.5. (i) *There is a unique least dagger subobject $\mathcal{H}^\dagger \subset \mathcal{T}(M)^\dagger$ whose Amice extension contains ℓ .*

(ii) *The subobject of minimum slope in its Amice extension $\mathcal{H}$ is ℓ .*

Proof. The ambient tensor is a candidate. Choose a candidate of least rank. Intersections of two candidates are candidates, since scalar extension is flat and the categories of (ϕ, ∇) -modules over the two fields are abelian. Minimal rank therefore implies that this candidate is contained in every other one and is unique.

Restriction of functionals to ℓ gives a map

$$\eta : (\mathcal{H}^\dagger)^\vee \longrightarrow \ell^\vee$$

which is $\mathcal{E}^\dagger$ -linear and compatible with Frobenius and connection; its target is an Amice module. To prove injectivity over $\mathcal{E}^\dagger$, we first verify that its kernel has invertible Frobenius linearization.

Both $\mathcal{E}^\dagger$ and $\mathcal{E}$ have basis $1, t, \dots, t^{p-1}$ over their respective Frobenius images. Indeed, group the exponents of a series modulo p , then apply inverse Witt Frobenius to its coefficients. The boundedness and overconvergence conditions survive this regrouping. It follows that

$$(6.6) \quad \mathcal{E}^\dagger \otimes_{\phi, \mathcal{E}^\dagger} \mathcal{E} \xrightarrow{\sim} \mathcal{E}, \quad a \otimes b \longmapsto a\phi(b).$$

Set $S = (\mathcal{H}^\dagger)^\vee$, $L = \ell^\vee$, and $Z = \ker \eta$, where the kernel is over $\mathcal{E}^\dagger$. Frobenius pullback over a field is flat. In the following diagram, the rows are exact at the terms containing Z and S :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \phi^* Z & \longrightarrow & \phi^* S & \longrightarrow & \phi^* L \\ & & \downarrow F_Z & & \downarrow F_S & & \downarrow F_L \\ 0 & \longrightarrow & Z & \longrightarrow & S & \longrightarrow & L. \end{array}$$

F_S is an isomorphism. Equation (6.6) identifies F_L with the Amice Frobenius linearization, which is also an isomorphism. Exactness of the rows therefore makes F_Z an isomorphism. Horizontality of η makes Z stable under the connection. If $Z \neq 0$, its annihilator in $\mathcal{H}^\dagger$ is a proper dagger subobject whose Amice extension still contains ℓ , contradicting minimality of $\mathcal{H}^\dagger$. Thus η is injective. Its image spans the rank-one Amice target.

Since $\ell^\vee$ is pure, [Tsu23, Theorem 2.14], which allows an arbitrary perfect residue field, shows that the maximal slope of $\mathcal{H}^\vee$ equals the slope of $\ell^\vee$ and that its maximal-slope quotient has the same rank as $\ell^\vee$. After dualizing, the subobject of minimum slope in $\mathcal{H}$ has slope $-r$ and rank one.

The Amice extension of η vanishes on every subobject of lower slope, so it factors through the canonical maximal-slope quotient of $\mathcal{H}^\vee$. The resulting map between rank-one objects is an isomorphism. Dualizing this factorization identifies the minimum-slope subobject of $\mathcal{H}$ with ℓ . $\square$

Proposition 6.6. (i) *The dagger hull of Proposition 6.5 is the scalar extension of a unique embedded good subobject $\mathcal{H}^+ \subset \mathcal{T}(M)$ over R .*

(ii) *The group H preserves $\mathcal{H}^+$.*

Proof. Pál's $\mathcal{E}^+$ is R , and his coefficient rings are defined over an arbitrary perfect field [Pal22, Definitions 2.1–2.2]. Since $\mathcal{T}(M)$ is defined over R , the subobject extension theorem, [Pal22, Theorem 2.14], extends $\mathcal{H}^\dagger$ to a good subobject. For two embedded good extensions, full faithfulness, [Pal22, Theorem 2.9], extends the identity on their common dagger extension to a unique good isomorphism compatible with their maps into $\mathcal{T}(M)$. The embedded extensions therefore coincide.

Each element of H preserves the good, dagger and Amice rings, the ambient tensor, and ℓ . It therefore permutes the dagger subobjects whose Amice extensions contain ℓ and preserves their least member, $\mathcal{H}^\dagger$. Uniqueness of its embedded good extension then gives invariance of $\mathcal{H}^+$ under H . $\square$

6.4. Saturation and uniqueness from the special fibre.

Lemma 6.7. (i) *Every finite R -module with connection is torsion-free.*

(ii) *If $L \subset P$ is an embedded good subobject of a finite free module, then P/L is finite free; consequently, the map $L/tL \rightarrow P/tP$ is injective.*

Proof. The ring R is a principal ideal domain: it is a one-dimensional localization of the regular local unique factorization domain $W(k)[[t]]$. Moreover it is differentially simple. To prove this, let $I = (f)$ be a nonzero differential ideal. Weierstrass preparation replaces its generator by a distinguished polynomial $P(t)$, up to a unit and a power of p . If $\deg P > 0$, then $P' \neq 0$ has smaller degree. Differential stability would imply $P' \in (P)$, whereas

$$R/(P) = K[t]/(P)$$

makes that impossible. If $\deg P = 0$, the ideal is the unit ideal.

Let N be a finite module with connection and let N_{tor} be its torsion submodule. If $fm = 0$, differentiating shows $f^2 \nabla(m) = 0$, so the connection preserves N_{tor} . Its annihilator is a nonzero differential ideal: if a annihilates it, the Leibniz rule and stability give $a'm = 0$ for every torsion element m . Differential simplicity implies $N_{\text{tor}} = 0$. The cokernel P/L inherits a connection, and is therefore torsion-free, hence free over the principal ideal domain. Tensoring $0 \rightarrow L \rightarrow P \rightarrow P/L \rightarrow 0$ with $R/(t)$ proves injectivity of $L/tL \rightarrow P/tP$. $\square$

Lemma 6.8. *Let P be finite free over R with connection. If two saturated submodules L_1, L_2 preserved by the connection have the same embedded special subspace in P/tP , then $L_1 = L_2$.*

Proof. The quotients P/L_i are free. Since R is a principal ideal domain, the torsion-free R -algebra $K[[t]]$ is flat. Over this larger ring the horizontal matrix equation with initial value the identity has a unique solution in $\mathrm{GL}_{\mathrm{rank} P}(K[[t]])$: the coefficient recursion divides by positive integers, which are invertible in K , and defines a formal power series over K .

Solving the restricted horizontal equation in each L_i identifies its base change with the constant submodule determined by its special subspace. Uniqueness for the ambient horizontal equation makes these identifications compatible with $L_i \subset P$. Thus the two submodules become equal over $K[[t]]$, and their spans over $K((t))$ are equal. Faithful scalar extension of vector spaces from $\mathrm{Frac}(R)$ to $K((t))$ gives equality of their spans over $\mathrm{Frac}(R)$. Finally, a saturated submodule is the intersection of its span over $\mathrm{Frac}(R)$ with P , so $L_1 = L_2$. $\square$

6.5. A rational natural projector. Lemma 6.7 gives an embedding

$$W = \mathcal{H}^+ / t\mathcal{H}^+ \subset \mathcal{T}(K^n)$$

in the marked special fibre.

Proposition 6.9. (i) *There is a GL_n -stable $\mathbf{Q}_p$ -subspace $W_0 \subset \mathcal{T}(\mathbf{Q}_p^n)$ with $W = W_0 \otimes_{\mathbf{Q}_p} K$.*
(ii) *There is a rational GL_n -equivariant idempotent e with image W_0 .*
(iii) *The natural action of e on $\mathcal{T}(M)$ satisfies*

$$\mathcal{H}^+ = \mathrm{im}(e_M).$$

Proof. Let $\rho_{\mathcal{T}}$ denote the rational representation of GL_n on $\mathcal{T}(\mathbf{Q}_p^n)$. Scalars act trivially, so the Honda identity in Theorem 6.1 gives

$$F_{\mathcal{T}}^n = \rho_{\mathcal{T}}(pI_n)\sigma^n = \sigma^n.$$

Since Frobenius preserves W , one has $\sigma^n(W) = W$. Let B_W be the unique reduced row-echelon matrix whose rows span W in the marked coordinates. Applying σ^n preserves the pivot positions and row-echelon conditions, so uniqueness gives

$$\sigma^n(B_W) = B_W, \quad (B_W)_{ij} \in K^{\sigma^n} = \mathbb{Q}_p^n.$$

The fixed-field equality follows by applying Frobenius to Witt coordinates and using $\{a \in k : a^{p^n} = a\} = \mathbb{F}_{p^n}$. The coefficient action γ_g fixes $\mathbb{Q}_p^n$, hence $\gamma_g(W) = W$. The comparisons preserve W by Proposition 6.6, and therefore

$$(6.7) \quad \rho_{\mathcal{T}}(R(g))W = \rho_{\mathcal{T}}(R(g))\gamma_g(W) = W \quad (g \in H).$$

An open subgroup of $\mathrm{SL}_1(D_n)(\mathbf{Q}_p)$ is Zariski dense in that connected algebraic group: in a smooth local chart a polynomial vanishing on a p -adic open subset vanishes identically. Density persists after finite extension of the ground field, by expanding coefficients in a rational basis. The marked Honda representation identifies the scalar extension of $\mathrm{SL}_1(D_n)$ to $\mathbb{Q}_p^n$ with SL_n . Thus W is SL_n -stable. Scalars act trivially on the tensor, so W is GL_n -stable as well.

Frobenius acts as $\rho_{\mathcal{T}}(A_0)\sigma$, where $A_0 \in \mathrm{GL}_n(\mathbf{Q}_p)$. The GL_n -invariance of W gives

$$\sigma(W) = \rho_{\mathcal{T}}(A_0)^{-1}F_{\mathcal{T}}(W) = W, \quad \sigma(B_W) = B_W.$$

Thus every entry of B_W belongs to $K^{\sigma} = \mathbf{Q}_p$. Let W_0 be its row space over $\mathbf{Q}_p$. This proves (i).

Complete reducibility of rational representations of GL_n in characteristic zero gives a GL_n -equivariant projection

$$e : \mathcal{T}(\mathbf{Q}_p^n) \longrightarrow \mathcal{T}(\mathbf{Q}_p^n), \quad e^2 = e, \quad \mathrm{im}(e) = W_0.$$

As an identity of rational representations, equivariance gives

$$(6.8) \quad e\rho_{\mathcal{T}}(B) = \rho_{\mathcal{T}}(B)e \quad (B \in \mathrm{GL}_n(D), D \text{ a } \mathbf{Q}_p\text{-algebra}),$$

where scalar extensions of e are denoted by the same symbol. Consequently the matrix e defines an operator e_M on $\mathcal{T}(M)$ independent of the chosen basis of M .

In such a basis write Frobenius and connection on M as $F_M = \Phi\phi$ and $\nabla_M = d + \Theta dt$, with $\Phi \in \mathrm{GL}_n(R)$ and $\Theta \in M_n(R)$. On the tensor module they are

$$F_{\mathcal{T}(M)} = \rho_{\mathcal{T}}(\Phi)\phi, \quad \nabla_{\mathcal{T}(M)} = d + d\rho_{\mathcal{T}}(\Theta) dt,$$

where $d\rho_{\mathcal{T}}$ is the induced Lie algebra representation. The rational matrix e satisfies $\phi(e) = e$ and $de = 0$. Applying (6.8) to $B = \Phi$ gives

$$F_{\mathcal{T}(M)}e_M = \rho_{\mathcal{T}}(\Phi)\phi(e)\phi = e\rho_{\mathcal{T}}(\Phi)\phi = e_M F_{\mathcal{T}(M)}.$$

Applying (6.8) to $B = I_n + \epsilon\Theta$ over $R[\epsilon]/(\epsilon^2)$ gives $[e, d\rho_{\mathcal{T}}(\Theta)] = 0$. Hence

$$\nabla_{\mathcal{T}(M)}(e_M v) = (e_M \otimes 1)\nabla_{\mathcal{T}(M)}(v).$$

Both the image and kernel of e_M are preserved by Frobenius and connection. Frobenius linearization restricts to an isomorphism on each summand. Thus $N = \mathrm{im}(e_M)$ is a good direct summand, with special fibre $W_0 \otimes_{\mathbf{Q}_p} K = W$. By Lemma 6.7, $\mathcal{H}^+$ is saturated. Lemma 6.8 applied to N and $\mathcal{H}^+$ gives $N = \mathcal{H}^+$, proving (iii). $\square$

6.6. Exclusion of the decomposable line.

Proof of Theorem 6.1. Suppose that Y exists. Form the line ℓ of (6.5) and the subobjects $\mathcal{H}^\dagger$ and $\mathcal{H}^+$ of Propositions 6.5 and 6.6. Work over $\mathcal{E}$ and choose a splitting of (6.3) in the category of $\mathcal{E}$ -vector spaces. This identifies $M_{\mathcal{E}}$ with $M_{\mathrm{et}} \oplus C$ as a vector space.

The natural equivariant projector of Proposition 6.9 makes $\mathcal{H}$ stable under every linear change of basis of $M_{\mathcal{E}}$, in particular under the block Levi subgroup $\mathrm{GL}(M_{\mathrm{et}}) \times \mathrm{GL}(C)$. The subobject $\mathcal{T}_{\min} = \bigwedge^s \mathrm{Hom}(C, M_{\mathrm{et}})$ from Lemma 6.4 is also preserved by this Levi subgroup. The intersection $\mathcal{H} \cap \mathcal{T}_{\min}$ is pure of slope $-r$ and contains ℓ . By Proposition 6.5, the subobject of minimum slope in $\mathcal{H}$ is ℓ , so the intersection equals ℓ . Thus the Levi subgroup preserves ℓ .

Since ℓ is decomposable, its underlying subspace $Q^\vee$ is recovered by

$$Q^\vee = \{v \in V^\vee : v \wedge \ell = 0\}.$$

It follows that the Levi subgroup preserves $Q^\vee \subset V^\vee = M_{\mathrm{et}} \otimes C^\vee$. This representation of $\mathrm{GL}(M_{\mathrm{et}}) \times \mathrm{GL}(C)$ is irreducible over $\mathcal{E}$. Indeed, the diagonal tori isolate the weight spaces spanned by matrix units in any invariant subspace. From a nonzero weight vector, elementary matrices in the two factors generate every matrix unit. Thus a nonzero invariant subspace is the whole tensor product.

This contradicts $0 < \dim Q^\vee = hr < hd = \dim V^\vee$. $\square$

Remark 6.10. The group H may act nontrivially on the perfected residue field. For $d = 1$ the condition $0 < r < d$ is empty, so the exclusion theorem is vacuous. At the endpoints $r = 0$ and $r = d$, the determinant lines are $\bigwedge^0 V^\vee$ and $\bigwedge^{hd} V^\vee$, respectively, and both are preserved by $\mathrm{GL}(M_{\mathrm{et}}) \times \mathrm{GL}(C)$.

7. RIGIDITY ON THE GENERIC FIBRE

Write $R = R_n$, $O = W(k_n)$, and $K = O[1/p]$. The action of S_n fixes O . The assertion that the $\mathbb{G}_n$ -stable ideals of $R_n[1/p]$ are (0) and $R_n[1/p]$ is attributed to Chai in [BHN22, Remark 4.2]. Lemma 7.3 extends to arbitrary height the Lie-theoretic method of [BHN22, first proof of Proposition 4.18].

Lemma 7.1. *Every maximal ideal of $R[1/p]$ is the kernel of evaluation at a point $\alpha \in \mathfrak{m}_L^{n-1}$ for a finite extension L/K .*

Proof. Let P be the contraction of the maximal ideal and set $A = R/P$. Then A is a complete local domain, p is a nonzero nonunit in A , and $A[1/p]$ is a field. Set $a = \dim A$. Extend p to a system of parameters $p, t_1, \dots, t_{a-1}$ and consider the continuous map

$$S = O[[T_1, \dots, T_{a-1}]] \longrightarrow A, \quad T_i \longmapsto t_i.$$

This map is finite. Choose lifts of a k_n -basis of $A/(p, t_1, \dots, t_{a-1})$. Successive approximation expresses every element of A as a linear combination of these lifts with coefficients in S . The parameter ideal defines the same topology as the maximal ideal, so completeness supplies the coefficients.

Let $J = \ker(S \rightarrow A)$. This ideal is prime, and finiteness gives $\dim(S/J) = \dim A = \dim S$. A regular local domain has only the zero prime at height zero. Thus $S \rightarrow A$ is injective. Since the field $A[1/p]$ is integral over $S[1/p]$, the ring $S[1/p]$ is a field. For $a > 1$, however, T_1 is a nonzero nonunit in $S[1/p]$. Hence $a = 1$ and A is finite over O .

The field $L = A[1/p]$ is finite over K , and integrality gives $A \subseteq O_L$. A power of the maximal ideal of A lies in pA , so each u_i has image in $\mathfrak{m}_L$. Continuity identifies $R \rightarrow A \subseteq O_L$ with evaluation of formal series at these images. $\square$

Lemma 7.2. *Let F/K be a complete discretely valued extension and let $\alpha \in \mathfrak{m}_F^{n-1}$. The homomorphism from $R[1/p]$ to the ring of analytic germs at α is injective.*

Proof. Translation by α defines a continuous substitution automorphism of $O_F[[u_1, \dots, u_{n-1}]]$, with inverse translation by $-\alpha$. Each translated coefficient is a convergent sum: the original coefficients are integral and the powers of α tend to zero. A bounded formal series defines an analytic function on every sufficiently small closed polydisc inside the open unit polydisc. If its germ vanishes, uniqueness of its power-series expansion makes every translated coefficient zero. Inverse translation then gives the zero series. Multiplication by a suitable power of p reduces every element of $R[1/p]$ to an integral series. $\square$

Let $\check{K} = \text{Frac } W(\overline{\mathbf{F}}_p)$ and let X be the Lubin–Tate open polydisc over $\check{K}$. Gross and Hopkins construct an equivariant etale period map

$$(7.1) \quad \Phi : X \longrightarrow \mathbf{P}_{\check{K}}^{n-1}.$$

We identify the hyperplanes in Gross and Hopkins’s representation with lines in its dual. Thus the target carries the dual of the representation obtained by splitting D_n over $\check{K}$ [HG94, Section 1.1 and Theorem 1].

Lemma 7.3. *Let $H \subseteq S_n$ be open, let $F/\check{K}$ be finite, and let $z \in X(F)$. An analytic function on $X_F = X \times_{\check{K}} F$ that vanishes at every H -translate of z has zero germ at z .*

Proof. Regard $\ell = \Phi(z)$ as a line in F^n . The infinitesimal projective action is the surjection

$$\text{End}_F(F^n) \longrightarrow \text{Hom}_F(\ell, F^n/\ell).$$

The images of an integral $\mathbf{Z}_p$ -basis of $\mathcal{O}_{D_n}$ span $\text{Hom}_F(\ell, F^n/\ell)$, because $D_n \otimes_{\mathbf{Q}_p} F \simeq M_n(F)$. Choose $b_1, \dots, b_{n-1}$ whose images form an F -basis. Etaleness of (7.1) gives a local analytic inverse ψ with $\psi(\ell) = z$. For sufficiently small parameters in F , define

$$(7.2) \quad \Omega(s_1, \dots, s_{n-1}) = \psi(\Phi(z) \cdot \exp(s_1 b_1) \cdots \exp(s_{n-1} b_{n-1})).$$

The derivative of Ω at zero is invertible, so the nonarchimedean inverse function theorem makes Ω an analytic chart after shrinking. Concretely, normalize its linear term to the identity and shrink the polydisc until the nonlinear remainder is a strict contraction. Iteration in the Tate algebra of this polydisc constructs the inverse over the complete field F .

Choose a sufficiently large that $1 + p^a \mathcal{O}_{D_n} \subseteq H$ and all exponentials with $s_i \in p^a \mathbf{Z}_p$ converge. At $p = 2$ take $a \geq 2$. Increase a further so that $z \cdot \exp(s_1 b_1) \cdots \exp(s_{n-1} b_{n-1})$ remains in the image of ψ for every $s_i \in p^a \mathbf{Z}_p$. This follows from continuity of the evaluated orbit: if $c < 1$ bounds $|p|$ and the absolute values of the coordinates of z , evaluation of an element of the N th power of the maximal ideal has absolute value at most c^N . Continuity of the marked action therefore implies continuity after evaluation at z .

On the product grid $(p^a \mathbf{Z}_p)^{n-1}$, equivariance and uniqueness of the inverse chart give

$$\Omega(s_1, \dots, s_{n-1}) = z \cdot \exp(s_1 b_1) \cdots \exp(s_{n-1} b_{n-1}) \in zH.$$

Let f be an analytic function on X_F vanishing on zH . Then $f \circ \Omega$ vanishes on $(p^a \mathbf{Z}_p)^{n-1}$. A one-variable analytic series vanishing at p^{a+j} for every $j \geq 0$ is zero near zero. Successive application in each variable, with the remaining variables fixed on their grids, shows that $f \circ \Omega$ has zero germ at zero. Since Ω is a chart, f has zero germ at z . $\square$

Theorem 7.4. *For every open subgroup $H \subseteq S_n$, the only H -stable ideals of $R[1/p]$ are (0) and $R[1/p]$.*

Proof. For $n = 1$, $R[1/p] = K$ is a field. Suppose $n > 1$ and let J be a nonzero proper H -stable ideal. Choose a maximal ideal containing J and let z be its evaluation point from Lemma 7.1. Choose a finite extension $F/\check{K}$ containing the coordinates of z , so that $z \in X(F)$. For $f \in J$ and $g \in H$,

$$f(z \cdot g) = (\rho_g f)(z) = 0.$$

By Lemma 7.3, the germ of f at z is zero; injectivity in Lemma 7.2 gives $f = 0$. This contradicts $J \neq 0$. $\square$

8. INVARIANT IDEALS AND THE TOPOLOGICAL COMPARISON

Lemma 8.1. *Let A be a complete Noetherian local ring with a continuous action of a compact group H . If an ideal J is H -stable, the stabilizer of each minimal prime of J is open in H .*

Proof. The group H permutes the finitely many minimal primes of J , so their stabilizers have finite index. A prime ideal P is closed in the topology defined by the maximal ideal. For generators $a_1, \dots, a_s$ of P , the condition $g(P) \subseteq P$ is the intersection of the closed conditions $g(a_i) \in P$. Imposing these conditions on both g and g^{-1} makes the stabilizer closed. A closed subgroup of finite index in a compact group is open, because its space of cosets is finite and Hausdorff. $\square$

Proof of Theorem 1.1. We first prove (ii). Let P be a proper prime stable under an open subgroup H of $\mathbb{G}_n$. If $p \notin P$, apply Theorem 7.4 to $P[1/p]$ and the open subgroup $H \cap S_n$ of S_n to obtain $P = 0$. If $p \in P$, choose the largest h with $I_h \subseteq P$. For $h = n$, maximality of I_n gives $P = I_n$. For $h < n$, the prime P/I_h of B_h avoids u_h . Theorem 5.1 gives $P/I_h = 0$, hence $P = I_h$. Conversely, marked isomorphisms preserve the scheme-theoretic height loci, so every I_h is stable.

For (i), let J be a proper H -stable radical ideal. By Lemma 8.1 and (ii), every minimal prime of J is one of $I_0, \dots, I_n$. These ideals are linearly ordered, whereas distinct minimal primes are incomparable. Thus J has a single minimal prime and equals it. The unit ideal I_{n+1} is also radical and stable. $\square$

Proof of Corollary 1.3. For (i), apply Corollary 5.2 to the open subgroup $H \cap S_n$ of S_n . The resulting identity has coefficients in B_h and group elements $g_i \in H \cap S_n \subseteq H$.

For (ii), let $0 \neq f \in R_n$ and define $J = \sum_{g \in H} R_n \rho_g(f)$. Theorem 7.4, applied to $H \cap S_n$, gives $J[1/p] = R_n[1/p]$. A finite expression for 1 and a common power of p clearing its denominators give the identity in (ii). $\square$

Let E_n be Morava E -theory. For $X \in \mathcal{C}_n$, define

$$M(X) = E_0^\vee X \oplus E_1^\vee X, \quad S(X) = \text{Supp}_{R_n} M(X).$$

The module $M(X)$ is finite over R_n and carries its semilinear $\mathbb{G}_n$ -action. Completed extension of scalars takes dualizable spectra to perfect E_n -modules and detects tensor-nilpotence [BHN22, Propositions 4.6–4.7]. The induced map

$$(8.1) \quad q : \text{Spec } R_n \twoheadrightarrow \text{Spc}(\mathcal{C}_n)$$

is surjective by [BHN22, Corollary 4.8], and the pullback of the Balmer support of X is $S(X)$ by [BHN22, Lemmas 4.4–4.5 and the proof of Corollary 4.8]. Consequently,

$$(8.2) \quad X \in \text{thick}_\otimes(Y) \iff S(X) \subseteq S(Y).$$

Indeed, surjectivity of (8.1) reflects inclusions of support. The criterion for generation by support is [BHN22, Corollary 2.4]. It applies because every thick tensor ideal in the rigid category $\mathcal{C}_n$ is radical: for $m \geq 1$, the object X is a retract of $X^{\otimes m} \otimes (DX)^{\otimes(m-1)}$, where DX denotes its dual.

Proof of Theorem 1.2. The radical of $\text{Ann}_{R_n} M(X)$ is invariant. Theorem 1.1(i) therefore assigns to each nonzero X a unique integer $t(X) \in \{0, \dots, n\}$ such that

$$\sqrt{\text{Ann}_{R_n} M(X)} = I_{t(X)}, \quad S(X) = V(I_{t(X)}).$$

Set $t(0) = n+1$. For a finite spectrum $F(j)$ of type j , the Morava module of $L_{K(n)}F(j)$ is I_j -torsion and is not I_{j+1} -torsion; its support is $V(I_j)$ [BHN22, Proposition 3.3 and the proof of Theorem 4.9]. Equation (8.2) gives

$$(8.3) \quad \mathcal{D}_j = \{X : t(X) \geq j\}.$$

The description (8.3) proves (ii), independence of the choice of $F(j)$.

For (i), let $\mathcal{J}$ be a nonzero thick tensor ideal. Choose $0 \neq Y \in \mathcal{J}$ for which $j = t(Y)$ is minimal. By (8.2), Y generates $\mathcal{D}_j$. Every object of $\mathcal{J}$ lies in $\mathcal{D}_j$ by (8.3), so $\mathcal{J} = \mathcal{D}_j$. The zero ideal is $\mathcal{D}_{n+1}$. Since $t(L_{K(n)}F(j)) = j$ for $0 \leq j \leq n$, all inclusions in (i) are strict.

For (iii), support of a tensor product is the intersection of supports, so

$$t(X \widehat{\otimes} Y) = \max\{t(X), t(Y)\}.$$

Together with (8.3), this formula proves that each $\mathcal{D}_j$, $1 \leq j \leq n+1$, is prime. Part (i) classifies all proper thick tensor ideals, so these are all the points of $\text{Spc}(\mathcal{C}_n)$. In the Balmer topology, the closure of a prime consists of the primes contained in it. Hence $\overline{\{\mathcal{D}_i\}} = \{\mathcal{D}_j : j \geq i\}$. $\square$

Remark 8.2. The support map (8.1) has the following description:

(i) For $\mathfrak{p} \in \text{Spec } R_n$,

$$q(\mathfrak{p}) = \mathcal{D}_{h(\mathfrak{p})+1}, \quad h(\mathfrak{p}) = \max\{h : I_h \subseteq \mathfrak{p}\}.$$

(ii) The fibre over $\mathcal{D}_{h+1}$ is $V(I_h) \setminus V(I_{h+1})$ for $0 \leq h \leq n$.

Indeed, [BHN22, Corollary 4.8] identifies $q(\mathfrak{p})$ with $\{X : M(X)_\mathfrak{p} = 0\}$, and (8.3) gives (i) and (ii). Under $2p-2 > n^2+n$, [BHN22, Theorem 4.13] proves the converse to [BHN22, Theorem 4.9]: the thick tensor ideal classification implies the classification of $\mathbb{G}_n$ -invariant radical ideals of R_n .

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The author prompted AI to try methods from the master’s thesis in exploring proof strategies. The author used AI for proof auditing, literature searches, reference verification and LaTeX formatting.

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